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Review article

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STORMING DATA IN THE CIRCULAR ECONOMY

循环经济中的风暴数据

Albana Gorishti ^{a,*}, Gloria Tyxhari ^a, Beris Ndreu ^b^a Faculty of Economy, University of Tirana
Tirana, Albania, albanagorishti@feut.edu.al, gloriatyxhari@feut.edu.al^b National Agency for Information Society
Tirana, Albania, berisndreu@gmail.com

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Abstract

The traditional linear approach of “manufacture, consume, discard” cannot be sustainable in a world with ever-expanding consumption rates. With regulations and the public becoming more and more sensitive toward environmental issues, companies seem pressured to shift to new business models. This work is focused on reviewing current technological solutions being applied in the transition to the circular economy. Focus will be on the opportunities created by applying real-time communication frameworks in business processes. This will be done by identifying current problems and solutions to those problems, improvements that can be made by introducing real-time communication, how real-time communication is engineered, and ways these improvements will lead to a more circular economy. This article explores the application of real-time communication frameworks, specifically leveraging Apache Storm, to optimize resource utilization, enhance sustainability, and facilitate the transition from a linear economy to a circular model.

Keywords: Real-Time Communication, Circular Economy, Sustainability, Apache Storm, Optimization

摘要 “制造、消费、丢弃”的传统线性方法在消费率不断扩大的世界中无法持续。随着法规和公众对环境问题越来越敏感，公司似乎被迫转向新的商业模式。这项工作的重点是审查当前应用于循环经济转型的技术解决方案。重点将放在通过在业务流程中应用实时通信框架所创造的机会。这将通过确定当前的问题和这些问题的解决方案、通过引入实时通信可以进行的改进、实时通信的设计方式以及这些改进将导致更多循环经济的方式来完成。本文探讨了实时通信框架的应用，特别是利用阿帕奇风暴，以优化资源利用、增强可持续性并促进从线性经济向循环模式的转变。

关键词: 实时通信、循环经济、可持续性、阿帕奇风暴、优化

I. INTRODUCTION

Several efforts have been made to address these industrialization drawbacks. A carbon emission footprint has been allocated to most products, industries are more conscious of their carbon emission, and some products have been restricted from usage. Awareness has increased in the population regarding everyday products that are used and the way they are manufactured and disposed [1]. Such measures have been put in place for a long time and have been shown to produce positive results regarding the environment. Although these actions have somewhat slowed down the degradation of the environment, there is more to be done in order to stop it, or even completely initiate a reverse process.

To meet the above-mentioned expectations, a transition from the traditional linear economy has to be made. Circular economy focuses on climate change, raw material scarcity, and resource usage efficiency. Recently, a lot of focus has been put toward it, and it has been proven to be helping in regards of what its focus is [2]. The circular economy is seen as a promising way to boost sustainability and competitiveness by decoupling economic growth from resource consumption and waste generation [3].

Even though there are many sectors moving toward a circular economy, one of its untackled points remains in technology. The implementation of circular economy principles is incentivized by digital technologies, which enable the adoption of new business models and the redesign of products and value chains to align with the paradigm of a smart circular economy [4]. Progress has been made; however, due to limitations in technology, this aspect can still be improved. One of the most important features that prevents the circular economy from progressing further in technology - related areas is the amount of data being transferred and the lack of computational efficiency available.

This paper will focus on improving the efficiency of processing data in operations related to a circular economy, mainly by introducing the effect of real-time data technologies in pushing the world further toward a circular economy, as well as analyzing the foundation of real-time technologies and their implementation on sectors where a circular economy can progress further.

II. METHODOLOGY

This section of the article outlines the approach used to conduct the review. To identify

relevant studies and technological solutions related to the transition to the circular economy, a comprehensive search was conducted. This involved searching various electronic databases, such as academic journals, conference proceedings, and industry reports. The search strategy involved using keywords and combinations of terms related to the circular economy, real-time communication, Apache Storm, sustainability, and optimization.

The articles and studies obtained from the search were then subjected to a selection process. Initially, duplicates were removed, and then the remaining articles were screened based on their titles and abstracts. The inclusion criteria were established to ensure that only studies related to technological solutions in the transition to the circular economy and the application of real-time communication frameworks in business processes were considered. The selected articles were then retrieved in full text for further analysis.

The selected articles were carefully reviewed and assessed for their relevance to the research questions and objectives of the review. The content of each article was examined to determine its contribution to understand the problems, solutions and improvements related to the circular economy and real-time communication. Only articles that provided substantial insights and relevant information were included in the review.

The information extracted from the selected articles was analyzed to identify current problems and solutions in the context of the circular economy. The focus was on understanding how real-time communication frameworks can be applied to business processes to enhance circularity. The analyses involved categorizing and summarizing the key findings, identifying patterns and trends, and critically evaluating the strengths and limitations of the existing technological solutions.

Overall, the methodology employed in this review article involved a systematic search, selection, and analysis process to gather relevant information and insights on the application of real-time communication frameworks in the transition to the circular economy.

III. RELATED WORK

Nowadays, with almost every industry preparing for the next industrial revolution, chances are for an enterprise to survive, it should be living on data. Today, the volume of information on the Internet increases

exponentially, especially that of interest for the media [5]. The Economist noted in 2017 that data value has for the first time surpassed that of oil. This trend is consistent with Statista projections that show enterprise spending on cloud and data centers increasing exponentially from 2009 to 2021, peaking at \$276.05 billion, with the total amount of data generated globally expected to reach 175 zettabytes by 2025.

Batch processing has been typical method under which firms have operated their processes. Usually batch processing includes a key/pair paradigm for both input and output of data. The process of computing and storing data is done through a master-slave architecture, where the master node distributes data across multiple slave nodes and is responsible for orchestrating data and spreading load [6].

Real-time analytics involves the analysis of data in the fastest possible manner, focusing on the most up-to-date information available [7]-[8]. This architecture has many benefits. Hadoop is a clear representative that meets the needs of business operations. It is horizontally scalable and designed for processing enormous amounts of data. [9].

However, this architecture has many downsides. One of them is the fact that this architecture is unable to perform recursive or iterative jobs/tasks. This means that the input should be full and ready before any of it is sent to the slaves for further processing or storage. Inherently, this combination cannot run continuous operations and queries. This is partly because the system does not maintain a state over previous or even future sessions [10].

This would greatly hinder the usability of circular economy-related sectors. An example would be the textile industry. Due to the slow processing times of stock and inventory databases, these industries are often left with excess stock. To maintain brand identity, excess stock (for example, seasonal clothing) is thrown away. Related research has shown that turnover can % [11].

On the other hand, real-time architecture uses scalable algorithms to compute, aggregate, and store large amounts of data in parallel. This is partly backward-compatible by using already in place master-slave architecture but using in-memory data. Storm would be a perfect representation of real-time processing and how it is causing the circular economy.

By using this technology in conjunction with other additional technologies, we can further improve data management, decision making, and efficiency and finally positively impact the

circular economy. Such technologies, on which a huge amount of interest has been shown recently, include IoT, Blockchain, and AI. We can mention real-time communication between devices on IoT [12], node syncing on Blockchain [9], task scheduling and parallel processing on AI [13]. The aforementioned technologies use real-time as their base implementation.

IV. REAL-TIME STORM

A. Real-Time Architecture

While real-time processing was being mentioned more and more as the key element missing for the transformation to the new business model, Apache Storm evolved with all the capabilities required to speed up conventional operations [14]. As described in “Mastering Apache Storm”, Apache Storm is a master-slave architectural design made of three components [15]-[19].

More specifically:

1) *Nimbus*

The Nimbus node acts as the master in a Storm cluster. It is responsible for spreading the application code across numerous worker nodes, assigning tasks to various machines, monitoring any errors, and restarting processes, as necessary. Nimbus is stateless and stores all its data in ZooKeeper. There is just one Nimbus node in a cluster of Storm nodes. The passive node will turn into an active node if the active node fails. The active Nimbus is designed to fail rapidly, so if it does, the passive node will take over as the active node or a restart can be performed in the down node with insignificant effect on the tasks now being completed on the worker nodes. In contrast, if the JobTracker breaks in Hadoop, the state of all running jobs will be unstable and a restart would be required. The Storm workers can continue to function normally even if all of the Nimbus nodes fail, but the user will not be able to add any new jobs to the cluster and it will be possible for the cluster to move the failed workers to another node [15], [18].

2) *Supervisors*

In a Storm cluster, worker nodes are known as supervisor nodes. Each supervisor node runs a supervisor daemon, which oversees spawning worker processes and initiating their start, stop, and execution to complete the tasks given to that node. A supervisor daemon is fail-fast, just like Nimbus, and keeps all of its states in ZooKeeper so that they can be restored after a restart without erasing any data. On that machine, a single supervisor daemon typically manages several worker processes [15], [19].

3) Zookeeper

Every distributed application needs some processes to interact with one another and share certain configuration information. An application called ZooKeeper offers each of these features in a trustworthy manner. Storm uses a ZooKeeper cluster as a coordinated application to manage numerous tasks. ZooKeeper serves as a central repository for all cluster-related statuses and Storm-related tasks. ZooKeeper is the way of communication for nimbus and supervisor since they do not interact directly. Because all data are stored in ZooKeeper, Nimbus and the supervisor daemons can both be quickly terminated without damaging the cluster.

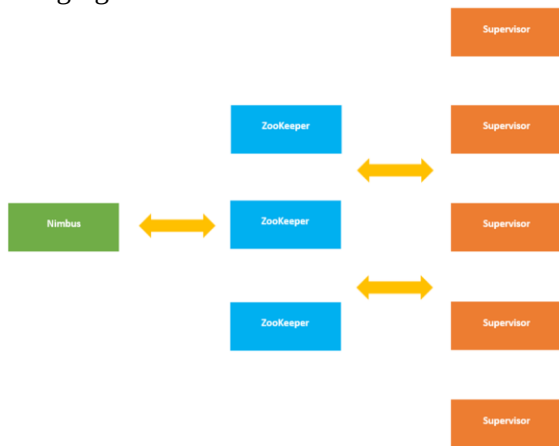


Figure 1. A typical Storm Cluster and its components (The authors' design)

B. Data Model

Tuples and streams make up the data model of real-time applications. More specifically, in a Storm data model:

1) Tuple

A single message or record that travels between different instances of a topology.

2) Stream

They serve as the core abstraction in the real-time ecosystem. An infinite series of tuples can be handled in parallel as a stream. A single bolt component or an array of bolt types can handle each stream. Streams are shown in Figure 2. by arrows. The bolts can make and consume tuples from each stream in a real-time application. Additionally, each stream also has an associated model for the tuples that it will process [15], [18]-[19].

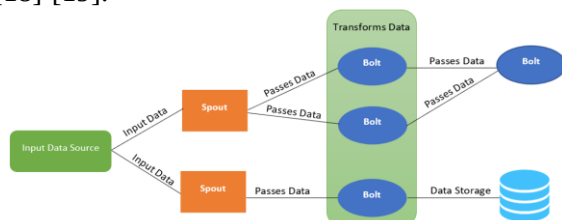


Figure 2. Real-time architecture Storm data model along with its building blocks (The authors' design)

C. Building Blocks

The logic of a real-time application is baked into the topology. The topology in Storm is made up of spouts and bolts.

1) Spouts

A spout is the source of tuples. It is in charge of reading data from external sources or listening for incoming messages in a queue, for example, and publishing that data—or emitting, in Storm terminology—into streams. A spout is capable of emitting many streams, each with a unique schema [19].

2) Bolts

A bolt transforms streams and is the central processing unit in a Storm topology. The tuples should be transformed just by one bolt in the topology, and many of these bolts can work together to reflect a complex transformation [15], [18]-[19].

V. REAL-TIME BACKED CIRCULARITY

Real-time communication is the backbone of most advances in recent technology, leading to the transition to a circular economy. Several prominent use cases can be found in various industries where real-time insights are actively supporting current decision-making, hence improving the performance of the circular economy. In this section, we provide an overview of how real-time data are driving the transition to a new business model.

Businesses, for instance, may avoid overproduction simply by establishing a solid reverse logistic infrastructure that enables a better product circulation. Throughout this process, real-time sensors may assess things for resale as well as return useful items to the stock inventory to cut down on waste [20]. Unusable products can be broken down into components for remanufacturing, or raw materials can be recovered and reused or recycled. Thanks to the application of these concepts, Intel is presently capable of reusing or retrieving 99% of its yields, which generated approximately USD 30 million in 2020, according to Supplychainbrain.

The retail business is also applying the same concept. Real-time data on consumer behavior allows retailers to estimate demand more effectively. Stock levels, which have been traditionally tracked manually using bar codes, are now accessible in real-time with blockchain tracking. Combined, these improvements enable businesses to provide customer service while reducing worldwide inventories by 20% [21].

On the other hand, to speed up and optimize

the flow of goods, smart warehouses, for instance, are already equipped with even more cutting-edge technology, namely robotic pickers, automated guided vehicles (AGVs), radio-frequency identification (RFID) tags and wearable electronics for warehouse workers. These devices are all connected via the Internet of Things [22].

Combining real-time monitoring of complex systems with 3D printing can remove the need to maintain significant stockpiles of replacement parts in physical warehouses. For example, sensors in line manufacturing machines can be used to quickly identify defective parts and print off a replacement nearby so that engineers can immediately install it. Accessing a virtual inventory of files and ready-to-assemble components enables this [23].

There are clear material advantages to manufacturing just the demanded parts, but there are other advantages derived from it. An airplane with a malfunctioning component can be fixed immediately upon landing, without having to wait for a new part to be installed. This implies that fewer vehicles are needed to conduct the service.

Material identification is another useful real-time data application in the circular economy. Even if it is feasible, manually sorting a range of material types into the right channels is time consuming and expensive due to the problem of dirty collection streams. Real-time data transmission technology-enabled sensor-enabled bins may sort recycling into the right streams by identifying, categorizing and crushing various things. This reduces waste and increases material recycling [24].

Moreover, real-time apps can support regenerative agricultural practices that proactively reconstruct biodiversity, improve soil health, and absorb carbon dioxide while maintaining the productivity of farmlands and reducing pressure to extend them where land use is necessary.

For example, rather than using physical "fences," cattle can be kept in designated regions using geographical "fences" that are configured on a smartphone app and communicate with solar-powered collars that each animal wears and that broadcast an audible warning when it leaves the designated area. Norwegian business Nofence is responsible for this grazing approach. Cows, goats, or many small animals are transferred continuously from pasture to pasture, maintaining soil organic matter and health as they move in a way that would be extremely labor-intensive with physical obstacles.

Furthermore, this type of technology has offered various ways for farmers to remotely control water use or completely automate the process by incorporating networked sensors to analyze farms, soil, and moisture [25]. Previous studies have shown that using too much irrigation and water results in the loss of roughly 80% of crops [26].

Regeneration has uses besides intelligent agriculture. In hopes of preventing the collapse of marine resources in their fisheries and enabling the natural ecosystem to rebuild alongside them, governments use real-time sensors and systems to monitor fishing boats. Furthermore, drones with real-time capabilities can quickly restore large tracts of forest by planting trees in deforested areas six times faster than human effort [27]-[28].

VI. CONCLUSION

This study indicates that the application of real-time technology plays a significant role in the transition from a linear economy to a more circular model. By leveraging real-time communication frameworks, businesses can effectively reduce raw material usage, enhance resource reusability, and minimize waste generation. The use of real-time technologies enables improved data collection and processing capabilities, providing businesses with better insights for resource management and waste reduction.

The implications of this study are twofold. First, it highlights the potential of real-time technology in facilitating the adoption of circular economy principles by businesses. By using real-time communication frameworks, companies can track products throughout their lifecycle, obtain accurate data, and make informed decisions optimize resource utilization. Second, the study emphasizes the significance of incorporating real-time capabilities into existing technologies or developing new technologies from scratch to bridge the gap between technology and circular economy objectives.

One strength of this study is its comprehensive review of existing literature and identification of current problems, solutions, and improvements related to the Circular Economy and real-time communication. The findings provide valuable insights into the benefits of real-time technology for achieving circularity. However, a limitation is that the study primarily focuses on technological solutions and may not encompass the broader social, economic, and policy aspects associated with transitioning to a circular economy. Future research should consider a more holistic approach that

incorporates these dimensions.

Based on this study, it is recommended that businesses consider integrating real-time communication frameworks into their operations to enhance resource management and promote circular practices. Further research should explore the practical implementation challenges and potential barriers to adopt real-time technologies in different industry sectors. Additionally, investigating the socio-economic implications and policy frameworks necessary to support the widespread adoption of real-time technology in the context of the circular economy would be valuable for future research endeavors.

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